Gautam Singh

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Education

Indian Institute of Science, Education and Research, Kolkata **June 2027**
5 Year BS-MS Dual Degree Programme

R. K. Mission Vidyapith, Deoghar, Jharkhand **June 2022**
AISSCE *90.2%*

R. K. Mission Vidyapith, Deoghar, Jharkhand **May 2020**
AISSE *96%*

Technical Skills

- **Languages** : Python, C, C++, Rust
- **Tools & Libraries** : LaTeX, Git, Linux, NumPy, scikit-learn, Pytorch, Tensorflow, CUDA

Honours and Awards

Reliance Foundation Undergraduate Scholarship **June 2023 – Present**
Batch of 2022

O.E.F Prize for Meritorious Students **Jan 2023**
For Exceptional Academic Performance in Class XII

O.E.F Prize for Meritorious Students **Jan 2021**
For Exceptional Academic Performance in Class X

Experience

Underwater AI **Sept 2025 – Present**
Founder

- Co-founded an AI startup focused on underwater imaging, formulating the problem as an inverse reconstruction task under severe noise, scattering, and color attenuation constraints.
- Developed and implemented deep learning and classical optimization-based models (PyTorch), combining priors from physics-based imaging models with data-driven approaches for image restoration.
- Built end-to-end ML pipelines for dataset creation, augmentation, and benchmarking; designed evaluation metrics for perceptual quality and downstream task performance (e.g., detection/segmentation robustness).
- Led research and technical direction, translating theoretical ideas from inverse problems and statistical learning into deployable prototypes; secured Rs. 5L grant through competitive technical pitching.

SlashDot: Programming Club **Sept 2023 – Aug 2024**
Club Leader

- Building a community of 450+ science and mathematics majors in using computational techniques in their everyday work.
- Orchestrating workshops, designed problems, coding sessions, and events to enhance skills and encourage scientific learning.
- Conducted programming contests and made problems on the Polygon platform.

Research Projects

Experiments on the Spanning Ratios of Convex Polygons **May 2024 – July 2024**
Project Supervisor: Prof. Pinaki Mitra, Indian Institute of Technology, Guwahati

- Read and implemented various research papers related to Geometric Spanner Networks.
- Wrote a codebase in C++ and Make to generate random polygons and calculate their spanning ratios.
- Implemented graph algorithms like Dijkstra's algorithm and Bellman Ford to solve the single-source and the multiple-source least path problems.
- Implemented a Dynamic Programming algorithm to solve the minimum weight triangulation problem.

Identification of figures of speech with LLM Models on Indic Languages **May 2025 – July 2025**

- Read and implemented various research papers related to LLM Finetuning.
- Wrote a codebase in Python to finetune several LLM models on Indic Language Datasets.
- Implemented several finetuning techniques like PEFT, RLHF, Prompt Tuning and Adapters.
- Implemented eval harness to accurately assess LLM performance.